

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023- 2024 (Odd Semester)

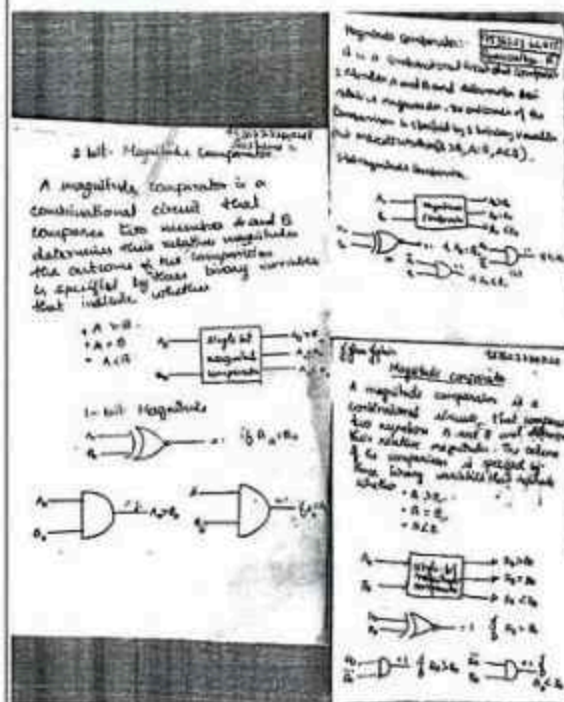
Active Learning Practices

Degree, Semester & Branch: B.Tech, III Semester & CSBS
Course Code & Title: CS3351 Digital Principles and Computer Organization
Name of the Faculty member: Dr.E.Emerson Nithiyaraj
Date: 06.10.2023

Active Learning Practices Execution

UNIT I - COMBINATIONAL LOGIC

Activity: One Minute Paper
Topic: Magnitude Comparator



E. Enl
07/10/23
Signature of the faculty


HOD



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Degree, Semester & Branch: B.Tech, III Semester & CSBS
Course Code & Title: CS3351 Digital Principles and Computer Organization
Name of the Faculty member: Dr.E.Emerson Nithiyaraj
Date: 02.11.2023

Innovative Practice Description

- **Unit / Topic:** Unit II / Flip Flops (SR & D) Operation,
- **Course Outcome:** CO2
- **UO Mapping:** 2b
- **Activity Chosen:** Mind Map
- **Time Allotted for the Activity:** 50 minutes

Details of the Implementation:

The Mind Map active learning activity on the topic of "SR and D Flip-Flops" aimed to enhance participants' understanding of these fundamental components in digital electronics. The activity focused on creating visual representations (mind maps) to explore the concepts, characteristics, and applications of SR and D flip-flops.

1. **Introduction to Mind Mapping: (2 minutes)**
 - Participants were introduced to the concept of mind mapping and its benefits in visualizing interconnected ideas. The importance of creating a hierarchical structure and using keywords in mind maps was emphasized.
2. **Explanation of SR and D Flip-Flops: (15 minutes)**
 - Brief lectures and discussions were conducted to review the basic principles of SR and D flip-flops, covering concepts such as state tables, excitation tables, and characteristic equations.
3. **Mind Map Creation: (3 minutes)**
 - Participants were divided into small groups and provided with flip chart papers, markers, and sticky notes. Each group was assigned either SR or D flip-flop, and they were tasked with creating a comprehensive mind map highlighting key concepts, inputs, outputs, and applications.
4. **Collaborative Discussions: (10 minutes)**
 - Groups collaborated to ensure a comprehensive coverage of the assigned flip-flop type. Facilitators circulated among groups, encouraging discussions, answering questions, and providing guidance on structuring the mind maps effectively.
5. **Group Presentations: (15 minutes)**

- Each group presented their mind map to the entire class. Presentations included explanations of the relationships between different components, practical applications, and potential variations of SR and D flip-flops.
6. **Feedback and Q&A Session: (5 minutes)**
- A feedback session was conducted where participants shared their thoughts on the activity. A Q&A session followed, allowing participants to seek clarification and engage in further discussions.

Feedback and Observations:

Participants gained a visual understanding of the connections between various aspects of SR and D flip-flops, fostering a holistic comprehension of the topic. The group setting facilitated collaborative learning, allowing participants to benefit from diverse perspectives and insights. Mind maps emphasized the real-world applications of SR and D flip-flops, enabling participants to connect theoretical knowledge to practical scenarios. The interactive nature of the activity contributed to better retention of information, as participants actively engaged with the material. Participants provided positive feedback on the effectiveness of the mind map activity in making the concepts of SR and D flip-flops more tangible and accessible. The collaborative aspect was particularly praised for promoting a dynamic and inclusive learning environment.

- **Images / Screenshot of the practice:**




31/10/2023
Signature of Faculty Member


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Degree, Semester & Branch: B.Tech, III Semester & CSBS

Course Code & Title: CS3351 Digital Principles and Computer Organization

Name of the Faculty member: Dr.E.Emerson Nithiyaraj

Date: 02.11.2023

Innovative Practice Description

- **Unit / Topic:** Unit III / Von Neumann Architecture
- **Course Outcome:** CO3
- **UO Mapping:** 3a
- **Activity Chosen:** Write Pair Share
- **Time Allotted for the Activity:** 50 minutes

Details of the Implementation:

The objective of the Active Learning Module (ALM) session titled "Write Pair Share" was to enhance students' understanding of Von Neumann Architecture, a fundamental concept in computer science and architecture. The ALM session was conducted on 02/11/2023 in B3L02 and was facilitated by Dr.E.Emerson Nithiyaraj. The "Write Pair Share" technique was chosen to promote active engagement and collaborative learning among students.

1. Introduction to Von Neumann Architecture:
 - Definition and key components.
 - Historical context and significance in computer science.
2. Individual Writing (10 minutes):
 - Students were given a set of questions related to Von Neumann Architecture.
 - Each student had to independently write their responses, encouraging personal reflection and knowledge recall.
3. Pair Discussion (15 minutes):
 - Students were paired up to discuss their written responses.
 - Emphasis on comparing and contrasting individual perspectives.
 - Encouraged students to articulate their thoughts and question their peers.
4. Group Sharing (20 minutes):
 - Each pair shared their key insights with the entire class.
 - Facilitator encouraged open discussions, and students were prompted to ask questions or seek clarification.
5. Wrap-up and Q&A (5 minutes):
 - Summary of the main points covered during the session.

- Opportunity for students to ask questions and seek clarification on any aspect of Von Neumann Architecture.

Feedback and Observations:

Engagement: The "Write Pair Share" format fostered a high level of engagement among students. The individual writing phase allowed for personal reflection, while the subsequent pair and group discussions promoted collaborative learning.

Inclusivity: The session encouraged the participation of all students, providing a platform for both vocal and non-vocal learners to contribute.

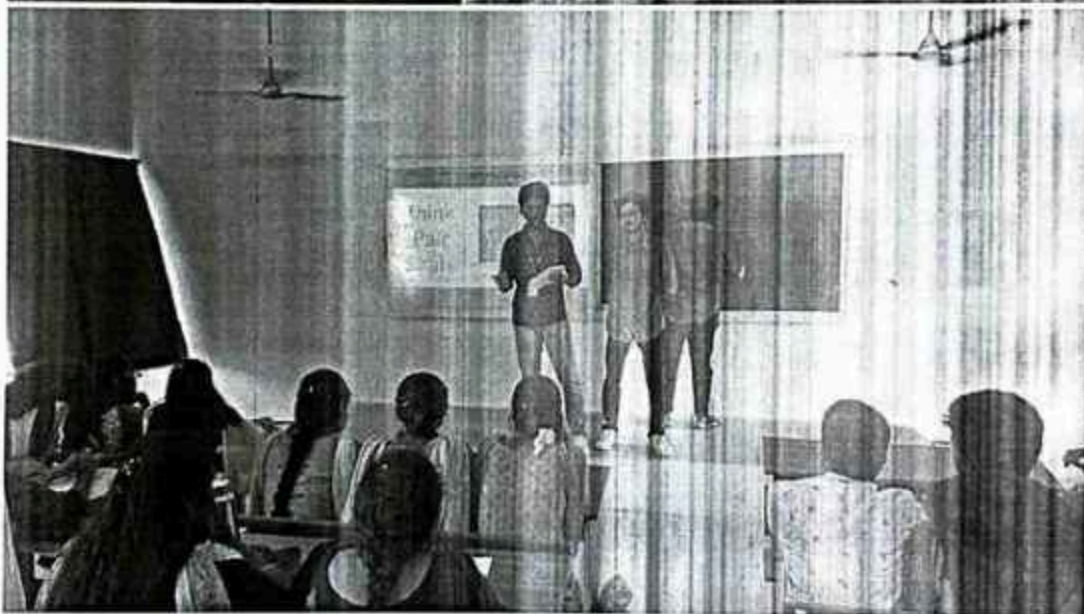
Concept Retention: The majority of students demonstrated a clear understanding of Von Neumann Architecture, as evidenced by their responses during the pair and group sharing phases.

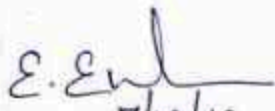
Facilitator's Role: The facilitator played a crucial role in guiding discussions, encouraging critical thinking, and ensuring that all students had the opportunity to express their thoughts.

The ALM "Write Pair Share" session on Von Neumann Architecture was successful in achieving its objective of deepening students' understanding of this foundational concept. The interactive nature of the session promoted active learning and peer-to-peer knowledge exchange. Further, the session provided valuable insights into individual student comprehension and allowed for timely clarification of any misconceptions. Overall, the ALM "Write Pair Share" session was a positive and effective learning experience for students, contributing to a more comprehensive understanding of Von Neumann Architecture.

• **Images / Screenshot of the practice:**






 7/12/23
 Signature of Faculty Member


 HOD

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Name of the Faculty member: Dr.E.Emerson Nithiyaraj

Date: 02.12.2023

Active Learning Practices Execution

UNIT IV – PROCESSOR (CO4)

Activity: Exit Slips

Topic: Hazards and Data Hazards (UO:4c)

Question Snapshot

Exit Slip

1. Write three things that you learned in today's lesson.
2. Write one question you have about today's lesson.



Answer Snapshots

* Hazard

It is a delay made during execution of instructions.

* Data Hazard:

■ The Hazard which is occurred because of the data dependency.

* Control Hazard:

The Hazard which is occurred because of fetching delay of data.

What are the methods to reduce data Hazard?

953622244001
V. Akshat.

Exit Slip

2/12/23
Gyaneshwari Mahapatra
953622244012

1. Data hazards
→ delay in the dependency of instruction.
 2. Instruction hazards
→ delay in the availability of instruction.
 3. Cache
→ temporary storage. Decode stage.
 4. Branch penalty can be reduced by starting at a
 5. Types of Hazards → data, instruction, structural.
 6. Instruction Hazard can be reduced by using queue.
+ Cache miss → it searches the data in the CPU memory.
- Question:

I don't have any questions in today's class sir.

Hazards.

Kamalkshi V
953622244022
2/12/23

Data Hazard
Instruction (or) Control Hazard
Structural Hazard.

Data Hazard:-

Dependency between the instruction

Control Hazard:-

delay during fetching instruction from the memory

Rectify the hazard by using the instruction queue.

By introducing hardware

Question:- Extra ~~extra~~ about the need concept about the topic

Observation:

A majority of students demonstrated a clear understanding of data hazards, accurately defining the concept and providing relevant examples. Responses regarding prevention strategies showed a range of understanding. Some students highlighted types of hazards, while others focused on cache miss and optimization techniques. Some students exhibited confusion between data hazards and control hazards, indicating a need for further clarification on the distinctions between these concepts. A few responses suggested a limited awareness of compiler optimization techniques for addressing data hazards. This indicates an area where additional explanation and examples may be beneficial. Thus, the Exit Slip served as a valuable tool for assessing student comprehension of data hazards, highlighting areas of strength and potential areas for further instruction.

E. E. Mahapatra
7/12/23
Signature of the faculty

HOD

**Department of Electronics and Communication Engineering**
Academic Year 2023 – 2024 (Odd Semester)**Degree, Semester & Branch: B.Tech., IIIrd Semester & CSBS****Course Code & Title: CS3351 Digital Principles and Computer Organization****Name of the Faculty member: Dr.E.Emerson Nithiyaraj, AP/ECE****Innovative Practice Description**

- **Unit / Topic:** Unit V / Cache Memories

- **Course Outcome:** CO5

- **UO Mapping:** 5b

- **Activity Chosen:** Jigsaw

- **Justification:**

Jigsaw is a powerful active learning strategy that encourages students to become responsible for their own learning while also fostering teamwork and cooperation. It is particularly effective for complex topics and subjects that benefit from multiple perspectives and insights. It is often used in educational settings to promote collaborative learning and deeper understanding of complex subject matter.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

- **Materials for the Activity:**

The materials for the preparation of the students are shared one week before through LMS Canvas. Including the material, students are asked to refer to the following book and web content also.

Book Title/ Author/ Publisher/ Edition	Page No.
1. David A.Patterson & John L.Hennessy, Computer Organization and Design, The Hardware/Software Interface, Elsevier, 2020.	383-417
Websites	
1. https://www.spiceworks.com/tech/tech-101/articles/what-is-cache/	
2. https://www.toppr.com/guides/computer-science/computer-fundamentals/primary-memory/cache-memory/	

- The topic on Cache Memories was subdivided into five sub categories. The Students are divided into groups of six students.
- Each group is allotted with a name and it is known as Expert group EG1, EG2, EG3, EG4 & EG5
- Each expert group is allotted with one topic
 - Cache Types
 - Cache Organization

- Cache Replacement Policies
- Cache Performance Optimization
- Cache in Modern Processors
- The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas. The Expert group members should go to other group and discuss the points (30 Minutes) to the other members.
- All the members in home group learnt about all the topics through expert group members. Finally one from each group should summarize the points of the topic Cache memories to the class (20 Minutes).
- Review of points and conclusion of the activity (5 minutes)
- Thus the concept of Cache Memories is known to all by involving the students actively.

• **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO2
CO5	H	H	M	M	M	M	L	L	L

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO 1	PO 9	PSO 1
	H	M	L
Justification for correlation	Students will be able to review and analyse and characterise the Cache memory. So, Course outcome is mapped at level H.	By studying the types of cache memory technologies based on a collaborative activity, Jigsaw helps the students present the results as a team, with smooth integration of contributions from all individual efforts. Hence, the outcome is mapped to level M.	The students will be able to identify the suitable memory chips for memory allocation. Thus, it is mapped at level L.

• **Key Learning:**

1. **Comprehensive Understanding:**

- Participants gained a comprehensive understanding of cache memories, covering various aspects such as types, organization, replacement policies, and coherency.

2. **Collaboration and Knowledge Sharing:**

- The jigsaw format facilitated collaboration and knowledge sharing among participants. It encouraged active engagement, and participants learned not only from their research but also from their peers' expertise.

3. Real-world Relevance:

- Discussions included real-world examples and applications of cache memories, highlighting their significance in improving computational efficiency and overall system performance.

4. Application to Course Content:

- Participants were able to connect their newfound knowledge to the course content, recognizing the practical implications of cache memories in computer architecture.

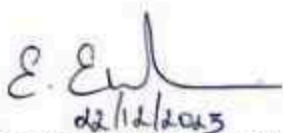
• Image / Screenshot of the practice:



Participants provided positive feedback on the effectiveness of the jigsaw activity in enhancing their understanding of cache memories. The hands-on approach, collaborative nature, and the opportunity to present and discuss findings were highlighted as valuable components of the activity.

References:

- David A. Patterson & John L. Hennessy, "Computer Organization and Design, The Hardware/Software Interface", Elsevier, 2020.
- John L. Hennessy and David A. Patterson, "Computer Architecture: A Quantitative Approach", Morgan Kaufmann, 2017.


02/12/2023

Signature of Faculty Member


HOD